

ZHIYANG ZHOU

zhou67@uwm.edu $\diamond$ <https://zhiyanggeezhou.github.io>

Assistant Professor of Biostatistics

University of Wisconsin-Milwaukee

METHODOLOGICAL RESEARCH TOPICS

Functional/longitudinal data analysis, deep learning, transfer learning, survival analysis, tensor data analysis, design of experiments.

COLLABORATIVE RESEARCH EXPERIENCE

I have collaborated with investigators across a wide range of research areas, including the atmospheric science, cardiovascular disease, diabetes, geriatric health, infectious diseases, mental health, and nutrition science.

EMPLOYMENT

University of Wisconsin-Milwaukee, U.S. Assistant Professor of Biostatistics (Tenure-track)	<i>2023–</i>
University of Manitoba, Canada Assistant Professor of Statistics (Tenure-track)	<i>2021–23</i>
Northwestern University, U.S. Research Scholar	<i>2020–21</i>
Simon Fraser University, Canada Sessional Instructor	<i>2019</i>

EDUCATION

Simon Fraser University, Canada Ph.D. in Statistics Supervisor: Richard Lockhart, Professor, Fellow of Royal Society of Canada	<i>2015–20</i>
Nankai University, China M.Sc. in Probability and Statistics Supervisor: Runchu Zhang, Professor	<i>2009–12</i>
Beijing Normal University, China B.Sc. in Statistics	<i>2005–09</i>

FUNDING

Natural Sciences and Engineering Research Council of Canada (NSERC) Principal Investigator, NSERC Discovery Grants Individual Program	<i>2022–27</i>
Principal Investigator, NSERC Discovery Launch Supplements	<i>2022–23</i>

PUBLICATIONS

Peer-Reviewed

14. H. Golandouz, M. Doupe, **Z. Zhou**, & L. Lix. (2026). Evaluating accuracy and fairness of machine-learning osteoporosis case definitions in linked administrative data. *International Journal of Population Data Science* 11:5:3553. DOI:10.23889/ijpds.v11i5.3553; PMID:PMC13426580.
13. **Z. Zhou**, Y. Deng, L. Liu, H. Jiang, Y. Peng, X. Yang, Y. Zhao, H. Ning, N. Allen, J. Wilkins, K. Liu, D. Lloyd-Jones, & L. Zhao. (2025). Deep neural network with a smooth monotonic output layer for dynamic risk prediction. *Statistics in Medicine* 45:e70401. DOI:10.1002/sim.70401; PMID:41640287; PMID:PMC12873558.
12. J. Liu, **Z. Zhou**, X. Cheng, D. Zhang, L. Li, & N. Vangeepuram. (2024). Food insecurity trends and disparities according to immigration status in the US households, 2011–2021. *Preventive Medicine* 187:108121. DOI:10.1016/j.ypmed.2024.108121; PMID:39208951; PMID:PMC12056756.
11. J. Liu, **Z. Zhou**, X. Cheng, & N. Vangeepuram. (2023). Geographic and sociodemographic variations in prevalence of mental health symptoms among US youths, 2022. *American Journal of Public Health* 113:1116–1119. DOI:10.2105/AJPH.2023.307355; PMID:37672739; PMID:PMC10484142.
10. Y. Deng, L. Liu, H. Jiang, Y. Peng, Y. Wei, **Z. Zhou**, Y. Zhong, Y. Zhao, X. Yang, J. Yu, Z. Lu, A. Kho, H. Ning, N. B. Allen, J. T. Wilkins, K. Liu, D. M. Lloyd-Jones, & L. Zhao (2023). Comparison of state-of-the-art neural network survival models with the pooled cohort equations for cardiovascular disease risk prediction. *BMC Medical Research Methodology* 23:22. DOI:10.1186/s12874-022-01829-w; PMID:36694118; PMID:PMC9872364.
9. **Z. Zhou** & P. Sang (2022). Continuum centroid classifier for functional data. *Canadian Journal of Statistics* 50:200–220. DOI:10.1002/cjs.11624.
8. Y. Zhao, Y. Wang, J. Liu, H. Xia, Z. Xu, Q. Hong, **Z. Zhou**, & L. Petzold (2021). Empirical quantitative analysis of COVID-19 forecasting models. *2021 International Conference on Data Mining Workshops (ICDMW)* 517–526. (Best Paper Award) DOI:10.1109/ICDMW53433.2021.00069.
7. **Z. Zhou** (2021). Fast implementation of partial least squares for function-on-function regression. *Journal of Multivariate Analysis* 185:104769. DOI:10.1016/j.jmva.2021.104769.
6. **Z. Zhou** (2019). Functional continuum regression. *Journal of Multivariate Analysis* 173:328–346. DOI:10.1016/j.jmva.2019.03.006.
5. **Z. Zhou** & R. Zhang (2014). A generalized general minimum lower order confounding criterion for nonregular designs. *Journal of Statistical Planning & Inference* 148:95–100. DOI:10.1016/j.jspi.2013.12.003.
4. W. Wang, D. Gong, **Z. Zhou**, & Y. Guo (2012). Robustness of the aerosol weekly cycle over Southeastern China. *Atmospheric Environment* 61:409–418. DOI:10.1016/j.atmosenv.2012.07.029.

Under Review

3. **Z. Zhou** & L. Zhao. (2025). Parsimonious joint model of survival outcome and multiple longitudinal risk factors.

Preprint

2. Y. Yang, & **Z. Zhou** (2025). Transfer learning for function linear regression via control variates. arXiv:2601.17217
1. **Z. Zhou**, & R. Lockhart. (2020). Partial least squares for sparsely observed curves with measurement errors. arXiv:2003.11542

SERVICE

Editorial Board Member

hLife (2024–28), *Journal of Multivariate Analysis* (2024–26)

Conference Committee Member

Scientific Committee for the International Applied Statistics Conference (UYIK, 2024–26)

Program Committee for ICSA-Canada Chapter 2024 Symposium

Grant Reviewer

NSERC Discovery Grants

Journal Reviewer

American Journal of Public Health, Annals of Statistics, Biometrics, Biostatistics & Epidemiology, Chemometrics and Intelligent Laboratory Systems, Computational Statistics, Canadian Journal of Statistics, Environmental Modeling & Assessment, hLife, IISE Transactions, Journal of Alzheimer's Disease, Journal of Applied Statistics, Journal of Computational and Graphical Statistics, Journal of Medical Internet Research, Journal of Multivariate Analysis, Journal of Statistical Theory and Practice, Knowledge-Based Systems, Mathematical Modelling and Analysis, Statistical Methods in Medical Research, Statistics and Computing, Statistics in Biosciences, Statistics in Medicine, Statistics & Probability Letters, Technometrics, The American Statistician, Trials

Conference Reviewer

Conference on Neural Information Processing Systems (NeurIPS), International Conference on Learning Representations (ICLR)

Committee Member @uwm.edu

Codification Committee (2026–28), Non-Academic Misconduct Hearing Committee (Dean of Students Office, 2025–26)

Committee Member @manitoba.ca

Graduate Committee (Dept. Stat., 2022–23), IT Committee (Dept. Stat., 2022–23), Research & Seminar Committee (Dept. Stat., 2021–22), Statistical Consulting Committee (Dept. Stat., 2022–23), Executive Committee (Faculty of Science, 2022–23)

Others

Official Nominator for the VinFuture Prize

INVITED PRESENTATIONS

2027: WNAR/IMS Annual Meeting; Lifetime Data Science (LiDS) Conference.

2026: ICSA China Conference.

2025: Joint Conference on Statistics and Data Science in China; WNAR/IMS Annual Meeting.

2024: International Joint Conference on Computational and Financial Econometrics (CFE) and Computational and Methodological Statistics (CMStatistics), CFE-CMStatistics 2024; International Conference on Econometrics and Statistics (EcoSta 2024); Joint Conference on Statistics and Data Science in China; SFU Statistical Learning for Large Scale Data Conference; ICSA-Canada Chapter Symposium; Southeast Wisconsin Data Science (SEAWINDS) Collaborative First Annual Research Symposium.

2023: International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2023); ICSA International Conference.

2022: International Conference of the ERCIM WG on Computational and Methodological Statistics (CMStatistics 2022); Biostatistics and Bioinformatics Round, George & Fay Yee Centre for Healthcare Innovation; Biostatistics Seminar Series, Northwestern University; ICSA-Canada Chapter Symposium; Machine Learning Special Interest Group Meeting, George & Fay Yee Centre for Healthcare Innovation; ICSA Applied Statistics Symposium.

2021: NIC-ASA & ICSA Midwest Chapter Joint Fall Meeting; ICSA Applied Statistics Symposium.

TEACHING

Supervision

- **Doctoral Students:** Sanaa Mohammed Sami Raja Mahmoud (2025–, Nursing, uwm.edu, with Dr. Jennifer Doering as the major supervisor), Donald Cramer (2023–, Biostatistics, uwm.edu).
- **Master Students:** Liliana Kasta (2024–25, Biostatistics, uwm.edu), Ian Nadolski (2023–25, Biostatistics, uwm.edu), Kelly Wikoff (2023–24, Biostatistics, uwm.edu) Ke Wang (2022–24, Statistics, umanitoba.ca), Yuting Kang (2023–25, Math, ctgu.edu.cn, with Drs. Chang-Lin Xiang and Changyu Guo as major supervisors).
- **Undergraduate Students:** Omar Hassan (2022–23, Physics & Astronomy, umanitoba.ca, with Dr. Wouter Deconinck as the major supervisor).

Courses

- **University of Wisconsin-Milwaukee:** Introduction to Biostatistics (2026), Data Analysis & Scientific Communication Incorporating Use of Generative AI (2026), Data Management and Visualization in R (2025–26), Probability and Statistical Inference (2024–25), Applied Survival Analysis (2024, 26).
- **University of Manitoba:** Multivariate Analysis (2022–23), Introduction to Statistical Inference (2022), Statistical Inference (2022).
- **Simon Fraser University:** Statistical Theory (2019).